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Case Title: Price Software DLP report

Case Overview

Title: Final Case - Price Software

Background: As sales manager, Tom is involved in all aspects of the company. He attends all the development meetings and has been with the company since its inception. As the company grows, plans have been made for its expansion. A new senior vice-president and two additional vice-president positions are to be created.

Everyone knows that Tom would like to be a vice-president.

In the past, Tom was involved in all aspects of the company. Recently, Tom was left out of key meetings where new promotions were discussed, which upset him.

Tom has made some statements recently that have concerned his co-workers. Some believe he isn't loyal to Price anymore and may be getting ready to leave the company. The current Management is concerned about the company's intellectual property.

Tom seems close with Leslie, the Finance Manager. They are often seen whispering at Leslie's desk, and there is a rumor that they are more than just colleagues

Objective: As the computer forensics analyst at Price Software, you are responsible for investigating the events and determining if there is any need for the company to be concerned about its intellectual property. If so, the results of your analysis will be used in any future employee disciplinary hearing. It is crucial to determine whether Tom has been doing anything inappropriate or if he is just angry about being left out of important decisions and maybe being passed over for a promotion. If there is any inappropriate behavior, it should be described in great detail.

Scope: This investigation is limited to the forensic analysis of the provided disk images using AccessData Forensic Toolkit (FTK) version 8.2 and X-Ways Forensics for RAID reconstruction. Antivirus software (AVAST) was also used to scan the systems to ensure no malicious files were present that could affect the integrity of the artifacts.

The evidence examined includes the Price Software Company file server image, the reconstructed RAID 5 array, and the individual workstation images belonging to Tom Warner and Leslie Stowle. No external devices or network captures were examined beyond those present in the provided disk images.

The analysis focuses on artifacts recovered from the disk images, including file system artifacts, browser activity, locally stored documents, email and communication artifacts, cloud storage artifacts, deleted files, and registry data. Particular attention was paid to identifying evidence of unauthorized access, data exfiltration, and any communications relevant to the company's concerns about intellectual property and employee conduct.

Methodology: To investigate potential unauthorized access and exfiltration of proprietary information at Price Software Company, standard digital forensic procedures were followed, beginning with documentation and preservation of evidence. Disk images of the company file server, RAID array, and the workstations belonging to Tom Warner and Leslie Stowle were provided for analysis.

The integrity of each forensic image was verified using MD5 hash values to ensure that the working copies matched the original evidence. Antivirus software (AVAST) was also used to scan each image to ensure no malicious software was present that could alter or generate artifacts.

Before analysis, the RAID 5 array was reconstructed from three physical disk images (RAID 5-1.E01, RAID 5-2.E01, RAID 5-3.E01) using X-Ways Forensics. The reconstructed array was then examined alongside the remaining disk images using AccessData Forensic Toolkit (FTK) version 8.2.

The first step of the examination was system identification, during which the operating system, file system, and user account structure of each examined system were determined. This established the foundation for artifact interpretation and user attribution across all systems.

Following system identification, analysis focused on multiple artifact categories across all examined systems, including browser history, locally stored and server-stored documents, downloaded files, deleted files, and Recycle Bin artifacts, communication artifacts, registry data, and stored credentials. Keyword searches were conducted within each forensic image using relevant terms to identify artifacts related to data access, exfiltration, and communication between users of interest.

All recovered artifacts were documented and correlated based on timestamps, file paths, and associated user accounts to reconstruct system activity and evaluate patterns of behavior across all examined systems. This correlation-based approach was used to assess the nature of the activity observed while maintaining a neutral and evidence-based interpretation throughout the investigation. Following the collection and analysis of all relevant artifacts, findings were organized into a chronological timeline to provide an efficient and clear understanding of the sequence of events as they relate to the concerns raised by Price Software Company management.

Executive Summary

Case Title: *Price Software*

Background: Price Software Company is a growing software development firm owned by Stayce Price. Tom Warner, the Senior Sales Manager, has been with the company since its inception and was widely expected to be considered for one of several newly created senior leadership positions. Recently, Tom was excluded from key meetings in which promotions were discussed, which visibly upset him. He has since made statements that concerned his co-workers, with some believing he may no longer be loyal to the company. Management has expressed concern regarding the potential exposure of the company's intellectual property. Tom has also been observed in frequent private conversations with Leslie Stowle, the Finance Manager, leading to speculation among staff about the nature of their relationship. Due to these concerns, a forensic examination of relevant company systems was initiated.

Objective: The objective of this investigation is to analyze the forensic images of Price Software Company's file server, RAID array, and the workstations belonging to Tom Warner and Leslie Stowle to determine whether there is evidence of unauthorized access, data exfiltration, or any other conduct detrimental to the company's interests.

The analysis focuses on identifying digital artifacts related to user activity, file access, communications, and the handling of proprietary information. All findings will be evaluated for relevance to a corporate disciplinary proceeding and documented accordingly.

Questions to be answered:

1. **What operating system and file system were in use across the examined systems, and how do they impact artifact interpretation and user attribution?**
2. **What digital artifacts link the observed activity to Tom Warner's user account, and how can that activity be attributed to him specifically?**
3. **Is there evidence that Tom Warner accessed, copied, or exfiltrated proprietary company data, including client lists, financial records, employee payment information, or account credentials?**
4. **What does Tom Warner's browser history, downloads, and locally stored documents reveal about his activities and intentions while using company systems?**

5. **Is there evidence of deleted files or Recycle Bin activity suggesting an attempt to conceal data access or removal?**
6. **What role, if any, did Leslie Stowle play in the events surrounding Tom Warner's activity, and is there evidence of collaboration or coordination between the two?**

Scope: This investigation is limited to the forensic analysis of the provided disk images using AccessData Forensic Toolkit (FTK) version 8.2 and X-Ways Forensics for RAID 5 reconstruction. No external devices or network captures were examined beyond those present in the provided disk images.

The analysis focuses on artifacts recovered from the file server, reconstructed RAID array, and the workstations belonging to Tom Warner and Leslie Stowle, including file system artifacts, browser activity, locally stored documents, communications, deleted files, and registry data. Particular attention was paid to identifying evidence of unauthorized access, data exfiltration, and any conduct relevant to the company's concerns about intellectual property and employee behavior.

Evidence: The evidence analyzed in this investigation consists of the following forensic images and associated data. The Price Software Company file server image (PSC Server OS.E01) was provided with an MD5 hash value of 520f7fc697b730ac11f43aded0380a7a. Tom Warner's workstation image (PSC Tom WS.E01) was provided with an MD5 hash value of d9d54d97378b090b7eac137986a26cdf. Leslie Stowle's workstation image (PSC Leslie WS.E01) was provided with an MD5 hash value of 8f6c72fa2e31efbe2296332044d86105. Additionally, three physical disk images comprising the RAID 5 array (RAID 5-1.E01, RAID 5-2.E01, RAID 5-3.E01) were provided with MD5 hash values of 3a5e0626f1241d620720225825a5d979, a7b327b9a12e7ba9c9087411da72b838, and df0231540e055e6f606a7ddb7044dc63, respectively.

All evidence was provided for analysis and was examined using forensic tools as outlined in the methodology. All forensic images were verified using MD5 hashing, and the calculated hash values matched the original acquisition values, confirming the integrity of the evidence.

Findings: The analysis revealed multiple important findings that help form a complete picture of the incident:

Hay.msg

Timestamp: 10/01/2004 6:08:38 AM

Hash: No separate MD5 because it is inside Outlook.ost

RE: Hay.msg

Timestamp: 10/04/2004 5:17:44 PM

Hash: No separate MD5 because it is inside Outlook.ost

Hotmail webmail cache artifact / dotattach389b7c[1].htm

MD5 shown in notes: d8561d4c1b430dd65b7574a985

Timestamp: Not clearly confirmed from the notes I can read

Company Credit Cards spreadsheet

Recovered from the reconstructed RAID 5 server image

Hash: No confirmed file hash

Start-up expenses spreadsheet / Carved [1244628].lnk

MD5: 4148c7ac77c6767d668618b3ddc872

Created: 09/30/2004 12:09:47 PM -0400

Last Write: 09/30/2004 12:16:00 PM -0400

Shortcut to Removable Disk (E).lnk

MD5: 73770be28d24ce87673d25ea4a23411

Created: 10/29/2004 14:10:08 UTC

Modified: 10/29/2004 14:10:08 UTC

Accessed: 01/03/2005

Dc1.lnk / Shortcut to c\$ on psc-ws-03 (H).lnk

MD5: 5aef7066c526123766526af6677f2a

Recycled: 10/01/2004 1:28:41 AM -0400

index.dat / Eraser download history artifact

MD5 from your screenshot: 971f961cf52d4575922b2e368f9b8aa

Accessed: 09/29/2004 2:41:10 PM -0400

Investigative Narrative: As the forensic analyst assigned to examine the digital evidence in the Price Software Company case, I conducted a structured forensic review of the disk images provided for analysis. The primary tool used during this investigation was AccessData Forensic Toolkit (FTK) version 8.2, which was used to analyze the forensic image files PSC Server OS.E01, PSC Tom WS.E01, and PSC Leslie WS.E01. X-Ways Forensics was used to reconstruct the RAID 5 array from the three physical disk images (RAID 5-1.E01, RAID 5-2.E01, RAID 5-3.E01) before analysis.

The examination focused on identifying artifacts related to user activity, locally stored documents, file access history, browser activity, communications, and deleted files across all examined systems. Particular attention was given to artifacts associated with the user account "twarner," including documents accessed or created on Tom Warner's workstation, files stored on the company file server and RAID array, and any communications or activity that may indicate unauthorized access to or exfiltration of company proprietary information.

The artifacts examined were evaluated based on file paths, timestamps, user directories, and associated metadata in order to reconstruct system activity across all examined systems. This process allowed for the identification of files that were intentionally created, accessed, or removed by specific users, rather than artifacts generated automatically through normal system operations.

The purpose of this investigation was to document the digital evidence present across the examined systems and evaluate whether the recovered artifacts reflect general user activity or conduct that may be detrimental to Price Software Company's interests. All observations in this report are based solely on artifacts recovered from the provided forensic images and are presented without assumption or speculation.

Phase 1: System Identification:

The first step in the examination was determining the operating systems, computer names, and user profile structures for the systems provided in the Price Software case. This was done by reviewing registry artifacts from the workstation and server images using FTK. The

SOFTWARE and SYSTEM registry hives were examined to identify the operating systems and computer names associated with each image.

Tom Warner's workstation was identified as a Microsoft Windows XP system with the computer name PC-WS-01. The primary user profile identified on the system was twarner, located under C:\Documents and Settings\twarner. This is important because artifacts recovered from this user profile, including Outlook data, Desktop shortcuts, browser artifacts, Recent files, and Recycle Bin artifacts, can be associated with Tom Warner's Windows profile.

Leslie Stowle's workstation was also identified as a Microsoft Windows XP system. The computer name recovered from the SYSTEM registry hive was PCS-WS-02. This separates Leslie's workstation from Tom Warner's workstation and helps distinguish artifacts recovered from each system during the investigation.

The Price Software Company server image was identified as a Microsoft Windows 2000 system with the computer name 2kADVserver. This confirms that the company server was separate from both user workstations and provides context for interpreting server-side file storage, finance shares, and RAID-related evidence.

During the examination, an additional referenced system name, psc-ws-03, was identified in a deleted shortcut from Tom Warner's workstation. This system name did not match Tom's workstation, Leslie's workstation, or the Price Software server image based on the system identification artifacts reviewed. Because of this, psc-ws-03 was treated as a referenced network system rather than one of the primary images examined in this report.

Purpose and What it Reveals:

System identification establishes the foundation for the rest of the investigation by showing which systems were examined, how they were configured, and how artifacts should be interpreted. Identifying Tom Warner's workstation as PC-WS-01 with the twarner profile supports attribution of user activity found under that profile. Identifying Leslie Stowle's workstation as PCS-WS-02 allows her activity to be separated from Tom's. Identifying the server as 2kADVserver supports later analysis of server-based financial records and RAID evidence.

This phase also helps prevent incorrect attribution. The separate psc-ws-03 reference shows that Tom's workstation contained a shortcut to another system's administrative share, but that system was not the same as Tom's workstation, Leslie's workstation, or the server image. This distinction is important because later findings involving network access must be interpreted carefully and supported by the available evidence.

Phase 2: User Attribution and Account Analysis

After identifying the systems, the next step was to determine which artifacts could be associated with specific user accounts. This phase focused on the user profiles found on Tom Warner's workstation and Leslie Stowle's workstation. On Tom Warner's workstation, the main user profile was identified as twarner, located at C:\Documents and Settings\twarner. On Leslie Stowle's workstation, the examined artifacts were associated with the Leslie workstation image and the lstowle user activity.

Tom Warner's user profile was significant because several important artifacts were located under the twarner profile path. These included Outlook data stored in C:\Documents and Settings\twarner\Local Settings\Application Data\Microsoft\Outlook\outlook.ost, a desktop shortcut to Removable Disk E:, and a deleted Recycle Bin shortcut referencing c\$ on psc-ws-03. Because these artifacts were found within Tom's user profile, they support attribution of the related activity to the twarner account.

Tom's Outlook data was especially important for user attribution. Several recovered email artifacts showed Tom communicating with Leslie Stowle about lunch, travel planning, and the vice president position. The emails titled "Vice Pres," "Hay," and "RE: Hay" were significant because they showed Tom's expectation of being promoted, his reaction after finding a memo stating he would not be promoted, and his belief that the company owed him. Since these messages were recovered from Tom's Outlook data, they help connect the communication activity directly to Tom Warner's account.

Tom's workstation also contained evidence of personal webmail activity. A Temporary Internet Files artifact showed Tom using the personal Hotmail account hotdog918@hotmail.com to contact Chris Melonis at an outside software company email address. The message referenced a meeting and stated that Tom would be able to let Melonis know "what we are doing next." This artifact is important because it shows Tom using a personal account outside the company email system, which is relevant to the investigation into possible communication with an outside software company.

Leslie Stowle's activity was also reviewed for attribution. Leslie's workstation was identified as PCS-WS-02, and her user activity included a carved shortcut artifact referencing \\2kadvserver\Finance\Spreadsheets\Start up expenses.xls. This artifact showed that Leslie's workstation accessed a financial spreadsheet located on the ADVserver finance share. Because Leslie was the Finance Manager, this activity is relevant to her job role, but it is also important to the investigation because it connects her workstation to sensitive company financial information.

The investigation also identified personal communication and travel-planning activity between Tom and Leslie. Recovered emails showed lunch, dinner, and Expedia travel or hotel links involving Los Angeles. These communications do not prove misconduct by themselves, but they support the conclusion that Tom and Leslie had a close personal relationship and communicated outside normal business matters. This relationship becomes more relevant when viewed with Tom's resentment about the promotion, his contact with Chris Melonis, and Leslie's access to financial records.

Purpose and What it Reveals:

The purpose of this phase is to connect digital activity to specific user accounts and profiles. In this case, artifacts located under C:\Documents and Settings\twarner\ support attribution of Outlook, desktop, webmail, removable media, and Recycle Bin activity to Tom Warner's user profile. This helps show that the most important communication and access artifacts were not random system files, but were tied to Tom's account activity.

This phase also shows Leslie Stowle's connection to the investigation. Her workstation accessed server-based financial information, and her communications with Tom show a close personal connection between the two users. While some of Leslie's activity may be consistent with her role as Finance Manager, it becomes important when correlated with the broader evidence, including Tom's motive, personal webmail activity, outside communication, and the presence of sensitive financial spreadsheets on the reconstructed RAID evidence.

Phase 3: Server and RAID Array Analysis

The next phase of the investigation focused on the Price Software Company server image and the reconstructed RAID 5 array. The server image was identified as Microsoft Windows 2000 with the computer name 2kADVserver. This was important because several financial and company records were associated with server-based storage rather than being stored only on Tom Warner's or Leslie Stowle's local workstation.

The RAID 5 array was reconstructed from the three physical RAID images using X-Ways Forensics before being reviewed with the other forensic evidence. The reconstructed RAID evidence contained company financial spreadsheets, including startup expense information and a company credit card spreadsheet. These files were important because they contained internal company financial information that would be sensitive in a business environment.

The startup expense spreadsheet contained internal financial planning information for Price Software Company. This included information related to owner investment, bank loan information, capital equipment costs, legal and accounting fees, salaries, and estimated startup capital. This type of information is relevant because it reflects internal business planning and financial data that would not normally be treated as public information.

A separate spreadsheet titled Company Credit Cards was also identified in the reconstructed RAID evidence. This spreadsheet contained employee names, job titles, credit card numbers, credit limits, and current balances for multiple Price Software employees, including Tom Warner and Leslie Stowle. This was significant because the spreadsheet contained sensitive company payment card information and showed that confidential financial records were present within the server or RAID evidence.

Leslie Stowle's workstation also contained a carved shortcut artifact referencing \\2kadvsrver\Finance\Spreadsheets\Start up expenses.xls. The shortcut was associated with network drive P: and showed that Leslie's workstation accessed the startup expense spreadsheet from the ADVserver finance share. This connection is important because it links Leslie's workstation activity to server-based financial files.

The server image also contained Internet history activity related to the download of Eraser, a secure deletion utility. The artifact was located under the Administrator profile in the Internet Explorer history path. The visible history showed access to Eraser57Setup.zip from SourceForge on 9/29/2004 at approximately 2:41 PM -0400. This does not prove that Eraser was installed or used to delete files, but it is relevant because the case involves deleted artifacts, possible data exposure, and concerns about concealment

Purpose and What it Reveals:

The purpose of this phase is to determine what sensitive company data was present on the server and reconstructed RAID evidence, and whether any workstation activity connected Tom or Leslie to that data. The reconstructed RAID evidence showed that financial records, including startup expenses and company credit card information, were present in the examined evidence. These records are relevant because they contained confidential business and payment information.

This phase also shows that Leslie's workstation accessed a financial spreadsheet from the 2kADVserver finance share. While Leslie's role as Finance Manager may explain some access to financial documents, the artifact is still important because it ties her workstation to sensitive company financial records during the same general period as Tom's promotion-related anger and communication with an outside software company.

The Eraser download history artifact adds possible anti-forensic context. It should not be overstated as proof that evidence was destroyed, but it does show that a secure deletion tool was accessed from the server environment. When viewed with the deleted shortcut artifacts, personal webmail activity, and sensitive company records, this server and RAID evidence supports the company's concern that internal information may have been placed at risk.

Phase 4: Browser History and Internet Activity Analysis

Phase 4 focused on browser history and internet-related artifacts recovered from the examined systems. This included reviewing Temporary Internet Files, Internet Explorer history artifacts, and cached webmail pages to identify activity related to Tom Warner, Leslie Stowle, and the Price Software server environment.

One of the most important internet artifacts was recovered from Tom Warner's workstation. A Temporary Internet Files artifact showed Tom using the personal Hotmail account hotdog918@hotmail.com. The cached webmail artifact showed a message to Chris Melonis at cmelonis@sapahsoftware.com. The message stated, "You should like this one. We have a meeting so I should be able to let you know what we are doing next. Cheers, TW." The artifact also showed an attachment named realarcade.exe. This is significant because it shows Tom communicating with an outside software company using personal webmail instead of company email.

The use of personal Hotmail is important because it may indicate an attempt to communicate outside normal company systems. The artifact does not prove by itself that Price Software intellectual property was sent, but the combination of personal webmail, an outside software company recipient, a meeting reference, and an executable attachment makes this finding highly relevant to the investigation.

Leslie Stowle's internet activity was also reviewed. The notes identified a cached Temporary Internet Files artifact related to "embezzlement" from Leslie's workstation. This artifact does not

prove that Leslie committed financial misconduct. However, it is relevant because Leslie was the Finance Manager, her workstation accessed a financial spreadsheet from the company server, and the reconstructed RAID evidence contained sensitive financial records including startup expense information and company credit card data.

Internet activity on the Price Software server was also reviewed. The server image contained Internet Explorer history activity related to downloading Eraser, a secure deletion utility. The history artifact was located under the Administrator profile in the Internet Explorer history path. The visible URLs referenced Eraser57Setup.zip from SourceForge and an EasyNews SourceForge mirror. The access activity occurred on 9/29/2004 at approximately 2:41 PM -0400.

The Eraser download history is relevant because Eraser is designed to securely delete files by overwriting data, making recovery more difficult. This finding does not prove that Eraser was installed, executed, or used to wipe specific files. However, it provides possible anti-forensic context because the investigation involves deleted artifacts, sensitive company information, and concerns about possible concealment.

Purpose and What it Reveals:

The purpose of this phase is to identify internet activity that may show communication, research, or possible preparation connected to the company's concerns. Tom Warner's Hotmail artifact is important because it shows communication with an outside software company through a personal account. This supports the concern that Tom may have been discussing company-related information outside official company channels.

Leslie's embezzlement-related internet artifact is relevant because it appears on the workstation of the Finance Manager and should be considered with the financial spreadsheet evidence. It should not be treated as proof of embezzlement by itself, but it adds context when viewed alongside Leslie's access to financial records.

The Eraser download history artifact provides possible anti-forensic context. While it does not prove that files were securely deleted, it shows that a file-wiping tool was accessed within the server environment. When combined with the other findings, this phase supports the broader concern that company information may have been accessed, discussed, or placed at risk through internet and webmail activity.

Phase 5: Document and File Analysis

Phase 5 focused on documents, spreadsheets, shortcuts, and file artifacts recovered from the workstation images, server image, and reconstructed RAID 5 evidence. This phase was used to identify whether sensitive company information was present in the evidence and whether any file artifacts connected Tom Warner or Leslie Stowle to that information.

The reconstructed RAID 5 evidence contained financial spreadsheet material related to Price Software Company. One spreadsheet titled Company Credit Cards contained employee names, job titles, credit card numbers, credit limits, and current balances. This was significant because it showed that sensitive company payment card information was present within the reconstructed RAID evidence. The spreadsheet included information for multiple Price Software employees, including Tom Warner and Leslie Stowle.

Additional financial spreadsheet evidence included startup expense information for Price Software Company. The startup expense documents contained internal business planning information, including owner investment, bank loan information, capital equipment costs, legal and accounting fees, salaries, and total estimated startup capital. This type of information is relevant because it reflects internal company financial planning and would be considered sensitive business information.

A carved shortcut artifact recovered from Leslie Stowle's workstation referenced the file \\2kadvsrver\Finance\Spreadsheets\Start up expenses.xls. The shortcut showed that the referenced file was located on the ADVserver finance share and was associated with network drive P:. The artifact recorded a creation time of 9/30/2004 at 12:09:47 PM -0400 and a last write time of 9/30/2004 at 12:16:00 PM -0400. This artifact is important because it shows Leslie's workstation accessed a financial spreadsheet from a server location rather than a file stored locally on her workstation.

Tom Warner's workstation also contained file artifacts relevant to the investigation. A shortcut to Removable Disk E: was located on the desktop of the twarner profile. The shortcut target information identified the local path as E:, with the volume type listed as Removable Disk and volume serial number D8A1-2398. The shortcut was created and modified on 10/29/2004 at 14:10:08 UTC. This artifact does not prove that files were copied, but it does show that Tom's profile had access to or interacted with removable storage.

Another important file artifact from Tom's workstation was the deleted shortcut named Dc1.lnk. This Recycle Bin artifact originally came from C:\Documents and Settings\twarner\My Documents\Shortcut to c\$ on 'psc-ws-03' (H).lnk. The shortcut referenced the administrative C\$ share on psc-ws-03 and appeared to be mapped or accessed as drive H:. This artifact is important because C\$ shares are commonly associated with administrative access to the root of a Windows system drive over the network.

The document and file evidence does not prove by itself that Tom copied company data or that Leslie misused financial information. However, the presence of sensitive financial spreadsheets, Leslie's access to a server-based finance spreadsheet, Tom's removable media shortcut, and Tom's deleted administrative share shortcut all provide important context for the investigation. These artifacts show that sensitive company records existed in the evidence and that user activity connected Tom and Leslie to file access paths that could be relevant to data movement or unauthorized access.

Purpose and What it Reveals:

The purpose of this phase is to identify important files and file-access artifacts that show what information was available, where it was stored, and which user activity connected to it. The reconstructed RAID evidence revealed sensitive company financial information, including credit card and startup expense spreadsheets. This supports the company's concern that internal business and financial records were present in the examined evidence.

This phase also helps connect Leslie Stowle's workstation to server-based financial data through the carved shortcut referencing the ADVserver finance share. While her role as Finance Manager may explain access to some financial records, the artifact is still relevant because it establishes a direct connection between her workstation and sensitive financial files.

Tom Warner's file artifacts are also important because they show access paths that could support data movement or unauthorized access. The removable disk shortcut shows interaction with removable media, while the deleted C\$ shortcut shows a reference to another system's administrative share. These findings do not prove exfiltration by themselves, but when viewed with Tom's motive emails, personal Hotmail communication, and outside software company contact, they support further concern about possible misuse of company information.

Phase 6: Deleted File and Recycle Bin Analysis

Phase 6 focused on deleted file artifacts and Recycle Bin activity recovered from the examined systems. This phase was important because deleted files and recycled shortcuts can show whether a user attempted to remove traces of file access, network access, or other activity relevant to the investigation.

A significant Recycle Bin artifact was identified on Tom Warner's workstation under the twarner user profile. The artifact was recorded as Dc1.lnk. The original file path showed that it came from C:\Documents and Settings\twarner\My Documents\Shortcut to c\$ on 'psc-ws-03' (H).lnk. The Recycle Bin metadata showed that the shortcut was recycled on 10/1/2004 at 1:28:41 AM -0400 and had not been removed from the Recycle Bin.

This shortcut is significant because it referenced the administrative C\$ share on psc-ws-03 and appeared to be mapped or accessed as drive H:. Administrative C\$ shares are commonly associated with access to the root of a Windows system drive over the network. The presence of this deleted shortcut does not prove that Tom copied files from psc-ws-03, but it does show that Tom's user profile had a shortcut referencing another system's administrative share.

The timing of this artifact is also important. The shortcut was recycled on 10/1/2004 at 1:28:41 AM -0400. Later that same morning, Tom sent the "Hay" email to Leslie Stowle stating that he found a memo saying he would not be promoted and that he would "get even." This timing does not prove that the deleted shortcut was related to the memo or any data movement, but it is relevant because it occurred during the same period as Tom's promotion-related frustration.

The investigation also identified a possible anti-forensic context through server's internet history showing activity related to downloading Eraser, a secure deletion utility. The Eraser-related artifact was an index.dat history artifact from the Administrator profile on the PSC Server OS image. The history showed access to Eraser57Setup.zip from SourceForge on 9/29/2004 at approximately 2:41 PM -0400. Eraser is designed to securely delete files by overwriting data, which can make recovery more difficult.

The Eraser artifact should be interpreted carefully. The history artifact shows download-related activity, but it does not prove that Eraser was installed, executed, or used to wipe specific files. To prove actual use, additional execution artifacts, installed program files, shortcut files, configuration files, logs, or wiping records would need to be found and correlated. Based on the evidence currently identified, the Eraser artifact should be treated as possible anti-forensic context rather than direct proof that evidence was destroyed.

Purpose and what it reveals:

The purpose of this phase is to determine whether deleted artifacts or Recycle Bin activity support the investigation's concerns about concealment, unauthorized access, or possible data movement. The deleted Dc1.lnk shortcut is important because it shows that Tom Warner's profile previously contained a shortcut to an administrative network share on psc-ws-03. Even though

the shortcut alone does not prove file copying, its deletion makes it relevant to the overall pattern of activity.

This phase also shows why deleted artifacts must be interpreted through correlation rather than assumption. The deleted shortcut, Tom's motive emails, removable media shortcut, and personal Hotmail communication each have separate meanings, but together they help establish a broader pattern of concern. The Eraser download history adds additional context because it shows that a secure deletion tool was accessed within the server environment. However, without proof of execution or wiping, it should only be described as possible anti-forensic activity.

Phase 7: Timeline Correlation

This phase was used to determine how Tom Warner's promotion-related emails, Leslie Stowle's financial access, Tom's outside communication, removable media artifacts, deleted shortcut activity, and server/RAID evidence fit together chronologically.

The timeline begins on 9/29/2004, when Internet history from the PSC Server OS image showed activity related to downloading Eraser, a secure deletion utility. The artifact was located under the Administrator profile and showed access to Eraser57Setup.zip from SourceForge at approximately 2:41 PM -0400. This does not prove that Eraser was executed, but it provides possible anti-forensic context early in the timeline.

On 9/30/2004, Leslie Stowle's workstation showed access to the server-based startup expense spreadsheet. A carved shortcut artifact referenced \\2kadvsrver\Finance\Spreadsheets\Start up expenses.xls and showed a creation time of 9/30/2004 at 12:09:47 PM -0400, with a last write time of 9/30/2004 at 12:16:00 PM -0400. Later that same day, Tom Warner emailed Leslie about the "MY NEW JOB" email and expressed excitement about the possible vice president position. This shows that financial access and promotion-related communication occurred close together in time.

On 10/1/2004 at 1:28:41 AM -0400, a deleted Recycle Bin shortcut from Tom Warner's profile showed a reference to c\$ on psc-ws-03 mapped as drive H:. This shortcut showed that Tom's profile had previously contained a shortcut to another system's administrative share. Later that morning, at 6:08:38 AM, Tom emailed Leslie and stated that he found a memo from Miss Price saying he would not be promoted to vice president. In that same message, Tom wrote that he

would “get even.” This makes 10/1/2004 a key date in the investigation because it contains both deleted network-share evidence and Tom’s clearest statement of anger and possible motive.

On 10/4/2004, the email activity continued. Leslie sent Tom Expedia travel or hotel links, and Tom replied that one of the links did not work. Later that day, Tom sent a follow-up email to Leslie stating that he saw the memo, had proof, worked at Price Software for seven years, helped start the company, and believed that the company owed him. In the same message, Tom mentioned having a meeting with someone and suggested dinner with Leslie. This message is important because it expands on Tom’s resentment and introduces the idea that he had a meeting with another person.

Also on 10/4/2004, Tom’s activity showed additional travel planning and outside communication. Tom had an email artifact involving an Expedia URL for a Los Angeles hotel or package search. The same general period also included the Hotmail webmail artifact showing Tom using the personal account hotdog918@hotmail.com to contact Chris Melonis at an outside software company. The message referenced a meeting and stated that Tom would be able to let Melonis know “what we are doing next.” The artifact also showed an attachment named realarcade.exe. This is significant because it connects Tom’s resentment and meeting reference to communication with an outside software-company contact through personal webmail.

On 10/29/2004, Tom’s desktop contained a shortcut to Removable Disk E:. The shortcut was created and modified on 10/29/2004 at 14:10:08 UTC and later showed an accessed date of 1/3/2005. This artifact does not prove that files were copied to removable media, but it shows that removable storage was referenced from Tom’s profile after the earlier promotion-related and outside communication activity.

When these artifacts are viewed together, the timeline shows a progression of concern. Tom first expected the vice president promotion, then learned he may not be promoted, then expressed anger and stated that he would “get even.” Shortly after, he continued stating that the company owed him and referenced a meeting. The Hotmail artifact then showed communication with Chris Melonis at an outside software company, including an executable attachment. At the same time, Leslie’s workstation was connected to financial spreadsheet access, and the reconstructed RAID evidence contained sensitive company financial documents, including startup expense and credit card spreadsheet information.

What it shows:

The purpose of this phase is to connect the individual artifacts into a chronological sequence so the activity can be understood as a pattern rather than isolated findings. The timeline shows that Tom Warner’s anger about the promotion decision occurred close in time to the deletion of network-share evidence, personal communications with Leslie, and later outside webmail communication.

This phase also shows how Leslie's activity fits into the broader investigation. Leslie's workstation accessed a server-based startup expense spreadsheet, and her emails with Tom show personal communication and travel planning. While these facts do not prove that Leslie helped Tom misuse company information, they make her activity relevant when viewed alongside Tom's motive, outside communication, and the financial records recovered from the RAID evidence.

Overall, the timeline supports the conclusion that Price Software had reason to be concerned about internal information being placed at risk. The evidence does not prove every possible allegation by itself, but the sequence of artifacts shows motive, access to sensitive financial information, outside communication through personal webmail, deleted network-share evidence, removable media activity, and possible anti-forensic context. This correlation gives the findings stronger meaning than they would have if reviewed separately.

Timeline Analysis

Date / Time	Artifact / Evidence	Source	Relevance
09/01/2004 6:17:26 PM	Lunch.msg	Tom Warner Outlook data	Tom emailed Leslie asking if she wanted to meet for lunch. This supports direct personal communication between Tom and Leslie.
09/29/2004 approx. 2:41 PM -0400	index.dat / Eraser download history artifact	PSC Server OS Administrator Internet History	Internet history showed activity related to downloading Eraser, a secure deletion utility. This provides possible anti-forensic context, but does not prove Eraser was executed.
09/30/2004 12:09:47 PM -0400	Start up expenses spreadsheet / Carved [1244628].lnk	PSC Leslie WS.E01	Leslie's workstation accessed \\2kadvserver\Finance\Spreadsheets\Start up expenses.xls from the server finance share.
09/30/2004 12:16:00 PM -0400	Start up expenses spreadsheet / Carved [1244628].lnk	PSC Leslie WS.E01	Last write time for the shortcut artifact showing Leslie's workstation connection to the server-based startup expense spreadsheet.
09/30/2004 5:54:39 PM	Vice Pres.msg	Tom Warner Outlook data	Tom emailed Leslie about the "MY NEW JOB" email and stated that he deserved the promotion and could use the extra money.
10/01/2004 1:28:41 AM -0400	Dc1.lnk / Shortcut to c\$ on psc-ws-03 (H).lnk	PSC Tom WS.E01 Recycle Bin	Deleted shortcut from Tom's profile referenced the administrative c\$ share on psc-ws-03, mapped as drive H:.
10/01/2004 6:08:38 AM	Hay.msg	Tom Warner Outlook data	Tom emailed Leslie stating that he found a memo saying he would not be promoted and wrote that he would "get even."
10/04/2004 10:30 AM	I like this one / Expedia travel email thread	Outlook email evidence	Leslie sent Tom an Expedia travel/hotel link. This supports personal communication and travel planning between Tom and Leslie.
10/04/2004 10:31 AM	RE: I like this one	Outlook email evidence	Tom replied that Leslie's travel link did not work.
10/04/2004 5:17:44 PM	RE: Hay.msg	Tom Warner Outlook data	Tom stated that he saw the memo, had proof, worked at Price Software for seven years, helped start the company, and believed the company owed him.
10/04/2004 5:23:56 PM	What do you think.msg	Tom Warner Outlook data	Tom emailed himself an Expedia URL involving Los Angeles hotel/package travel.
10/04/2004 5:31:39 PM	RE: I like this one.msg	Outlook email evidence	Leslie emailed Tom an Expedia URL involving Los Angeles hotel/package travel.
Date not confirmed in notes	Hotmail webmail cache artifact / dotattach389b7c[1].htm	PSC Tom WS.E01 Temporary Internet Files	Tom used hotdog91@hotmail.com to contact Chris Melonis at cmelonis@sapahsoftware.com, referenced a meeting, and included realarcade.exe.
Date not confirmed in notes	Company Credit Cards spreadsheet	Reconstructed RAID 5 server image	Spreadsheet contained employee names, job titles, credit card numbers, credit limits, and current balances.
10/29/2004 14:10:08 UTC	Shortcut to Removable Disk (E).lnk	PSC Tom WS.E01 Desktop	Shortcut to removable disk E: was created and modified under Tom's profile, showing removable media activity.
01/03/2005	Shortcut to Removable Disk (E).lnk	PSC Tom WS.E01 Desktop	Accessed date for the removable disk shortcut.

The timeline shows that the most important activity occurred within a short period between 9/29/2004 and 10/4/2004, with later removable media activity appearing on 10/29/2004. The sequence begins with server's internet history showing Eraser download activity, followed by Leslie Stowle's workstation accessing a server-based startup expenses spreadsheet. Shortly after that, Tom Warner emailed Leslie about his expected promotion and then later stated that he found a memo saying he would not be promoted and that he would "get even."

The timeline is important because it shows that Tom's anger about the promotion was not isolated from the rest of the evidence. The deleted shortcut to the administrative C\$ share on psc-ws-03 occurred during the same general period as Tom's promotion-related emails. Tom's later email stating that the company owed him also occurred close in time to travel-planning emails between Tom and Leslie and the Hotmail artifact showing Tom communicating with Chris Melonis at an outside software company.

When viewed together, the timeline supports the conclusion that Price Software Company had reason to be concerned. The evidence shows a progression from promotion-related frustration, to resentment, to outside communication through personal webmail, while sensitive company financial records were present in the server and RAID evidence. The timeline does not prove every allegation by itself, but it connects the separate artifacts into a pattern that supports further concern about possible misuse or exposure of company information.

Conclusion

This investigation identified multiple artifacts showing that Price Software Company had reason to be concerned about Tom Warner's activity and the possible exposure of company information. The strongest evidence came from Tom Warner's Outlook messages, which showed that he expected to be promoted to vice president, became upset after finding a memo stating he would not receive the position, and later wrote that he would "get even." In another message, Tom stated that he had proof, had worked at Price Software for seven years, helped start the company, and believed the company owed him.

The investigation also identified evidence that Tom used a personal Hotmail account to communicate with Chris Melonis at an outside software company. The recovered webmail artifact referenced a meeting and stated that Tom would be able to let Melonis know "what we are doing next." The same artifact showed an attachment named `realarcade.exe`. Although the evidence does not prove by itself that Price Software's intellectual property was sent, the use of personal webmail, the outside software company recipient, the meeting reference, and the

executable attachment are highly relevant to the company's concern about possible disclosure of internal information.

Leslie Stowle's activity was also relevant to the investigation. Her workstation accessed a startup expense spreadsheet from the ADVserver finance share, and the reconstructed RAID 5 evidence contained sensitive financial records, including startup expense information and a company credit card spreadsheet. The company credit card spreadsheet contained employee names, job titles, credit card numbers, credit limits, and current balances. While Leslie's role as Finance Manager may explain access to some financial information, these artifacts are still important because they show that sensitive company financial data was present and accessible within the examined environment.

The investigation also identified supporting artifacts related to possible data movement and concealment. Tom Warner's workstation contained a shortcut to Removable Disk E:, showing removable media activity associated with his profile. A deleted Recycle Bin shortcut also referenced the administrative C\$ share on psc-ws-03, mapped as drive H:. These artifacts do not prove that Tom copied files, but they are relevant because removable media and administrative shares can be used to access or move data. In addition, server internet history showed activity related to downloading Eraser, a secure deletion utility. This should be interpreted as possible anti-forensic context, not proof that files were wiped.

Overall, the evidence supports the conclusion that Tom Warner's activity was more than ordinary workplace frustration. His messages show motive and resentment, his personal webmail activity shows contact with an outside software company, and the surrounding file artifacts show access paths that could be relevant to data movement. Leslie Stowle's workstation and the reconstructed RAID evidence also show access to and storage of sensitive company financial information. While not every artifact proves misconduct by itself, the timeline and correlation of the evidence show a pattern of concern involving motive, sensitive company records, outside communication, removable media, deleted network-share evidence, and possible anti-forensic activity. Based on these findings, Price Software Company had a reasonable basis to be concerned about the protection of its intellectual property and internal financial information.

Item I-Tom finding out about the memo

From: Tom Warner
To: Leslie Stowle[lstowle@PSC.local]
Cc:
Date: Fri 10/1/2004 6:06:38 AM
Subject: Hay

You are not going to believe this.

I found a memo that miss Price wrote that says she is not going to promote me to vice president. I will get even. Who does she think she is. I am going home now

From	To	Cc	Subject	Date	Received	Submit Time	Last Modified	1	0	3
		All Employees	New Positions	8:59	9/9/2004 5:...			1	0	0
Leslie Stowle	Tom Warner		RE: I like this one	17,647	10/4/2004 5:...			1	0	0
Leslie Stowle	Tom Warner		I like this one	8,080	10/4/2004 5:...			1	0	0
Tom Warner	Leslie Stowle		Lunch	6,946	9/11/2004 6:17 PM			1	0	0
Tom Warner	Leslie Stowle		Vice Pres	8,226	9/9/2004 5:...			1	0	0
Tom Warner	Leslie Stowle		Hay	7,520	10/1/2004 6:...			1	0	0
Tom Warner	Leslie Stowle		RE: Hay	14,705	10/4/2004 5:...			1	0	0
Tom Warner	Tom Warner		What do you think	8,252	10/4/2004 5:...			1	0	0

(re local machine)						
	ID5	SHA1	SHA256	Created	Accessed	Modified
18	909e8af77dbd0c0f7e0eb073ec	F4eHb...		09/29/2004 17:06:46 (09/29/2004 21:06:46 UTC)	01/03/2005 17...	09/29/2004 17:06:46 (09/29/2004 21:06:46 UTC)
18	c3061ee58de65077fa40ba4fe12	a70S...		11/29/2004 12:52:54 (11/29/2004 17:52:54 UTC)	01/03/2005 17...	01/03/2005 17:02:31 (01/03/2005 22:02:31 UTC)
18	7628ab5e720425236ef9e2be0	ad0e...		11/29/2004 02:55:17 (11/29/2004 02:55:17 UTC)	01/03/2005 17...	01/03/2005 17:02:31 (01/03/2005 22:02:31 UTC)

*Item II - Chris Melonis emails***Hotmail webmail cache / dotattach — Tom emails Chris Melonis with realarcade.exe attached**

The screenshot shows a forensic analysis tool interface. The main window displays an email from Chris Melonis to Tom. The email content is as follows:

To: cmelonis@aspahsoftware.com
Cc:
Bcc:
Subject: This is a good one!
Attachments: realarcade.exe (0.20 MB)
You should like this one. We have a [plugin](#) to I should be able to let you know what we are doing next. Cheers, TW

The file list at the bottom shows the following files:

Name	Item #	Ext	Path	Category	HEX	SHA1
component.js	40189	js	PSC Server OS.E01[Partition 1]NOMAN [NTFS]root\Program Files\Enchanted\controls\component.js	7 bit text	b0e3d5212b35a4e476d4b767d7d79	f1953...
dotattach389967c711.htm	11497	htm	PSC Tom WS.E01[Partition 1]NOMAN [NTFS]root\Documents and Settings\homer\Local Settings\Temporary Internet Files\Cont...	HTML	955a1d6113435d955d7574a0d576	1bca47...
dsrec.dat	30544	dat	PSC Tom WS.E01[Partition 1]NOMAN [NTFS]root\WINDOWS\system32\dsrec.dat	7 bit text	b2b689487b3f49726d65e7d160d38	1e5d5...
dsrec.dat	31725	dat	PSC Server OS.E01[Partition 1]NOMAN [NTFS]root\WINDOWS\system32\dsrec.dat	7 bit text	b53300a0d6f9c4a07764f4e46d03	07b3e...
dsrec.dat	37121	dat	PSC Leale WS.E01[Partition 1]NOMAN [NTFS]root\WINDOWS\system32\dsrec.dat	7 bit text	b2b689487b3f49726d65e7d160d38	1e5d5...

The search results on the right show a list of items, including:

- Item 15903 [A0001948.rtf] PSC Tom WS.E01[Partition 1]NOMAN
- Item 33374 [dabtpc.rtf] PSC Tom WS.E01[Partition 1]NOMAN
- Item 31614 [dabtpc.rtf] PSC Leale WS.E01[Partition 1]NOMAN
- Item 83334 [Three_A.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 83337 [Three_F2.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 83338 [Three_F3.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 16025 [A0001948.rtf] PSC Leale WS.E01[Partition 1]NOMAN
- Item 7941 [Office XP Professional with FrontPage Setup(0003)...
- Item 8108 [Office XP Professional with FrontPage Setup(0003)...
- Item 11151 [UPDAT110.htm] PSC Tom WS.E01[Partition 1]NOMAN
- Item 17656 [SCHORESP.CFG] PSC Tom WS.E01[Partition 1]NOMAN
- Item 83100 [Three_C1.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 17660 [SCHORESP.CFG] PSC Tom WS.E01[Partition 1]NOMAN
- Item 17659 [SCHORESP.CFG] PSC Tom WS.E01[Partition 1]NOMAN
- Item 17658 [SCHORESP.CFG] PSC Tom WS.E01[Partition 1]NOMAN
- Item 17657 [SCHORESP.CFG] PSC Tom WS.E01[Partition 1]NOMAN
- Item 10928 [SCHORESP.CFG] PSC Leale WS.E01[Partition 1]NOMAN
- Item 10929 [SCHORESP.CFG] PSC Leale WS.E01[Partition 1]NOMAN
- Item 10930 [SCHORESP.CFG] PSC Leale WS.E01[Partition 1]NOMAN
- Item 10931 [SCHORESP.CFG] PSC Leale WS.E01[Partition 1]NOMAN
- Item 14604 [UPDAT110.htm] PSC Leale WS.E01[Partition 1]NOMAN
- Item 10927 [SCHORESP.CFG] PSC Leale WS.E01[Partition 1]NOMAN
- Item 83099 [Three_C2.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 83776 [Calendar-view.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 83445 [Calendar-view.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 83333 [Three_C1.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 80193 [calendar.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 80188 [component.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 17136 [dotattach389967c711.htm] PSC Tom WS.E01[Partition 1]NOMAN
- Item 15999 [dotattach389967c711.htm] PSC Tom WS.E01[Partition 1]NOMAN
- Item 83404 [Recur.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 80153 [webdata.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 76246 [E001.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 37121 [dsrec.dat] PSC Leale WS.E01[Partition 1]NOMAN
- Item 30544 [dsrec.dat] PSC Tom WS.E01[Partition 1]NOMAN
- Item 83775 [Component.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 83444 [Component.js] PSC Server OS.E01[Partition 1]NOMAN
- Item 75891 [dotattach389967c711.htm] PSC Server OS.E01[Partition 1]NOMAN
- Item 11496 [dotattach389967c711.htm] PSC Tom WS.E01[Partition 1]NOMAN
- Item 11497 [dotattach389967c711.htm] PSC Tom WS.E01[Partition 1]NOMAN

*Item III -Company cards found on the raid server***Company Credit Cards spreadsheet — Recovered from reco**

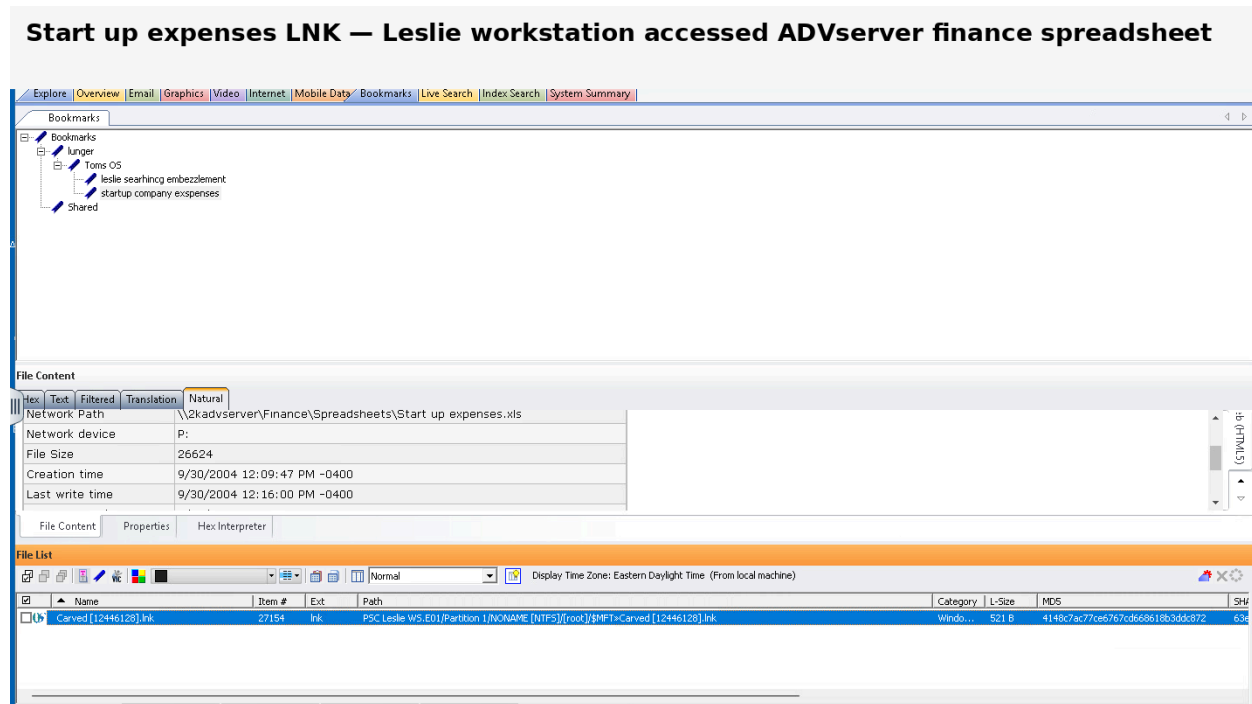
Raid 5-1	Raid 5-2	Raid 5-3	Raid 5-3, Raid2	Company Credit Cards.xls								
	A	B	C	D	E	F	G	H	I	J		
1	Company Credit Cards											
2	Name	Job Title	Card Number	Credit Limit	Curent Balance							
3	Stayce Price	President	3789-8745-8702-0005	\$50,000.00	\$2,678.00							
4	Larry Ezeerf	Vice President	3745-0347-0000-2371	\$25,000.00	\$0.00							
5	Tom Warner	Senior Sales Manager	3756-9234-8735-0023	\$25,000.00	\$3,760.00							
6	Leslie Stowle	Finance Manager	3798-0362-1111-9532	\$5,000.00	\$0.00							
7	Mike Relwof	Development Manager	3769-0023-2234-9116	\$25,000.00	\$1,769.00							
8	George Bushman	Sales Rep.	3756-9800-7689-0548	\$10,000.00	\$12,873.00							
9	Howard Fox	Sales Rep.	3789-4576-0234-1876	\$10,000.00	\$18,347.00							
10	Chuck Cobbhead	Sales Rep.	3798-0455-9812-3948	\$10,000.00	\$13,932.00							
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
21												
22												
23				\$160,000.00	\$53,359.00	\$106,641.00						
24												
25												
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37												
38												
39												

Item IV- business start-up expenses

Start-Up Capital Estimate spreadsheet — Recovere

Start-Up Capital Estimate				
Price Software Company				
	Monthly Expenses	Cash Needed to Start	% of Total	Source of Estimate
MONTHLY COSTS				
Salary of owner-manager	\$4,000	\$8,000	2.6%	
All other salaries and wages	25,000	75,000	24.3%	
Rent	7,968	23,904	7.7%	
Advertising	1,500	4,500	1.5%	
Delivery expense	2,400	7,200	2.3%	
Supplies	1,000	3,000	1.0%	
Telephone	900	2,700	0.9%	
Other utilities	1,350	4,050	1.3%	
Insurance	4,500	4,500	1.5%	
Taxes, including social security	6,723	26,892	8.7%	
Interest	2,381	7,143	2.3%	
Maintenance	578	1,734	0.6%	
Legal and other professional fees	3,923	11,769	3.8%	
Miscellaneous	5,000	15,000	4.9%	
Subtotal		\$195,392	63.3%	
ONE-TIME COSTS				
Fixtures and equipment		\$9,800	3.2%	
Decorating and remodeling		4,690	1.5%	
Installation charges		2,632	0.9%	
Starting inventory		5,967	1.9%	
Deposits with public utilities		5,387	1.7%	
Legal and other professional fees		50,000	16.2%	
Licenses and permits		3,491	1.1%	
Advertising and promotion for opening		2,653	0.9%	
Cash		25,000	8.1%	
Other		3,740	1.2%	
Subtotal		\$113,360	36.7%	
TOTAL ESTIMATED START-UP CAPITAL		\$308,752	100%	

Item V- uploading spreadsheets to a network drive, mostlikely th raid server



*Item VI- Removable disk short cut in recycler***Shortcut to Removable Disk (E).lnk — Tom profile removable media artifact**

The screenshot displays a forensic tool interface with a top menu bar (File, Edit, View, Evidence, Filter, Tools, Manage, Help) and a toolbar. The main window is divided into several panes. On the left, the 'Evidence Items' pane shows a tree view of the file system. The central pane, titled 'Shortcut File', displays 'Link target information' for a file named 'Shortcut to Removable Disk (E).lnk'. The information includes the local path, volume type, volume serial number, file size, and creation time. Below this, a 'File List' pane shows a table of files. The table has columns for Name, Label, Item #, Ext, Path, Category, P-Size, MD5, Created, Accessed, and Modified. The file 'Shortcut to Removable Disk (E).lnk' is listed with item # 9089 and a creation time of 10/29/2004 14:10:08.40. The bottom status bar shows 'Loaded: 1', 'Filtered: 1', 'Total: 1', 'Highlighted: 0', 'Checked: 0', and 'Click to update size'.

Link target information

Local Path	E:\
Volume Type	Removable Disk
Volume Serial Number	DBA1-2398
File Size	0
Creation time	1/1/1980 2:00:00 AM -0500

File List

Name	Label	Item #	Ext	Path	Category	P-Size	MD5	Created	Accessed	Modified
Shortcut to Removable Disk (E).lnk		9089	lnk	PSC Tom WS.ED1\Parti...	Windo...	184 B	73770ba2b8d4aca87673d25ea4a23411	10/29/2004 14:10:08.40 (10/29/2004 18:10:08.40 UTC)	01/03/2005 15:...	10/29/2004 14:10:08.40 (10/29/2004 18:10:08.40 UTC)

Loaded: 1 Filtered: 1 Total: 1 Highlighted: 0 Checked: 0 Click to update size

PSC Tom WS.ED1\Partition 1\NONAME [NTFS]\(root)\Documents and Settings\hwaner\Desktop\Shortcut to Removable Disk (E).lnk

Ready [Explore Tab Filter: None]

Item VII- Eraser download

index.dat / Eraser download history — PSC Server OS Administrator profile

Forensic Toolkit Version: 8.2.2.179 SP2 HF4 Database: localhost (Postgres) Case: Case 3 Final

File Edit View Evidence Filter Tools Manage Help

Filter: -unfiltered- Filter Manager Smart View

Explore Overview Email Graphics Video Internet Mobile Data Bookmarks Live Search Index Search System Summary

Bookmarks

- lunger
 - downloaded eraser(winner)
 - leddie email (sweet dog)
 - leddie email
 - rating on meetings
 - Tone OS
 - leddie seahing embeclement
 - startup company expenses
 - Shared

Bookmark Information

Bookmark Name: downloaded erase Creator Name: lunger

Comments Timeline Project VSC
File Edit Supplementary File

Be Content

Hex Text Filtered Translation Natural

use counts: 0

URL: :2004092920040930: Administrator@Host: prdownloads.sourceforge.net

File:

user name:

response:

accessed time: 9/29/2004 2:41:10 PM -0400

File Content Properties Hex Interpreter

Be List

Display Time Zone: Eastern Daylight Time (From local machine)

Name	Label	Item #	Ext	Path	Category	P-Size	L-Size	MD5	SHA1	SHA256
index.dat		7252	dat	PSC Server OS E01/Partition 1/NOName [NTFS]/(root)/Documents and Settings/Administrator/Local Settings/History/History IE5/M9-Hin012004092920040930/index.dat	PSC IE...	32,00 KB	32,00 KB	47079615F2564657592326a36878dab	6556C...	

Indexed: 1 Filtered: 1 Total: 1 Highlighted: 1 Checked: 0 Click to update size

PSC Server OS E01/Partition 1/NOName [NTFS]/(root)/Documents and Settings/Administrator/Local Settings/History/History IE5/M9-Hin012004092920040930/index.dat

ready [Bookmarks Tab Filter: None]

Item VIII-travel links back and forth between Tom and Leslie

[illegible]

Item IX – Expected promotion with Leslie

Vice Pres.msg – Tom discusses expected promotion with Leslie

Default Viewer [Vice Pres.msg]

From: Tom Warner

To: Leslie Stowle[stowle@PSC.local]

Cc:

Date: Thur 9/30/2004 5:54:39 PM

Subject: Vice Pres

Hay Leslie,

Did you see the e-mail about "MY NEW JOB" Wee I can't wait, god knows I deserve it and I sure can use the a-tra money.

PS. Want to do lunch. Hee Hee

HD

From	To	Cc	Subject	Size	Received	Submit Time	Last Modified		Alter
Stacy Price	All Employees		New Positions	8,183	9/30/2004 5:...			1 0 0 3	
Leslie Stowle	Tom Warner		RE: I like this one	17,547	10/4/2004 5:...			1 0 0 1	
Leslie Stowle	Tom Warner		I like this one	8,080	10/4/2004 5:...			1 0 0 1	
Tom Warner	Leslie Stowle		Lunch	6,946	9/1/2004 6:17 PM			1 0 0 3	
Tom Warner	Leslie Stowle		Vice Pres	6,226	9/30/2004 5:...			1 0 0 3	
Tom Warner	Leslie Stowle		Hay	7,520	10/1/2004 6:...			1 0 0 3	
Tom Warner	Leslie Stowle		RE: Hay	14,705	10/4/2004 5:...			1 0 0 3	
Tom Warner	Tom Warner		What do you think	8,357	10/4/2004 5:...			1 0 0 3	

From local machine

#	MD5	SHA1	SHA256	Created	Accessed	Modified
1	9099d5d77d5dc0f5bdcf0e80377ec	He4b...		09/29/2004 17:06:46.6 (09/29/2004 21:06:46.6 UTC)	01/03/2005 17:...	09/29/2004 17:06:46.6 (09/29/2004 21:06:46.6 UTC)
0 KB	230631e046a50776a43e0e149b172	a76355...		11/01/2004 12:52:29.54 (11/01/2004 17:52:29.54 UTC)	01/03/2005 17:...	01/03/2005 17:02:31.75 (01/03/2005 22:02:31.75 UTC)
1 KB	7626a1ba2716a29e2e2d3947eef2be0	a1d8e3...		11/29/2004 21:02:55.17 (11/30/2004 02:02:55.17 UTC)	01/03/2005 17:...	01/03/2005 17:02:31.75 (01/03/2005 22:02:31.75 UTC)

Folder: K

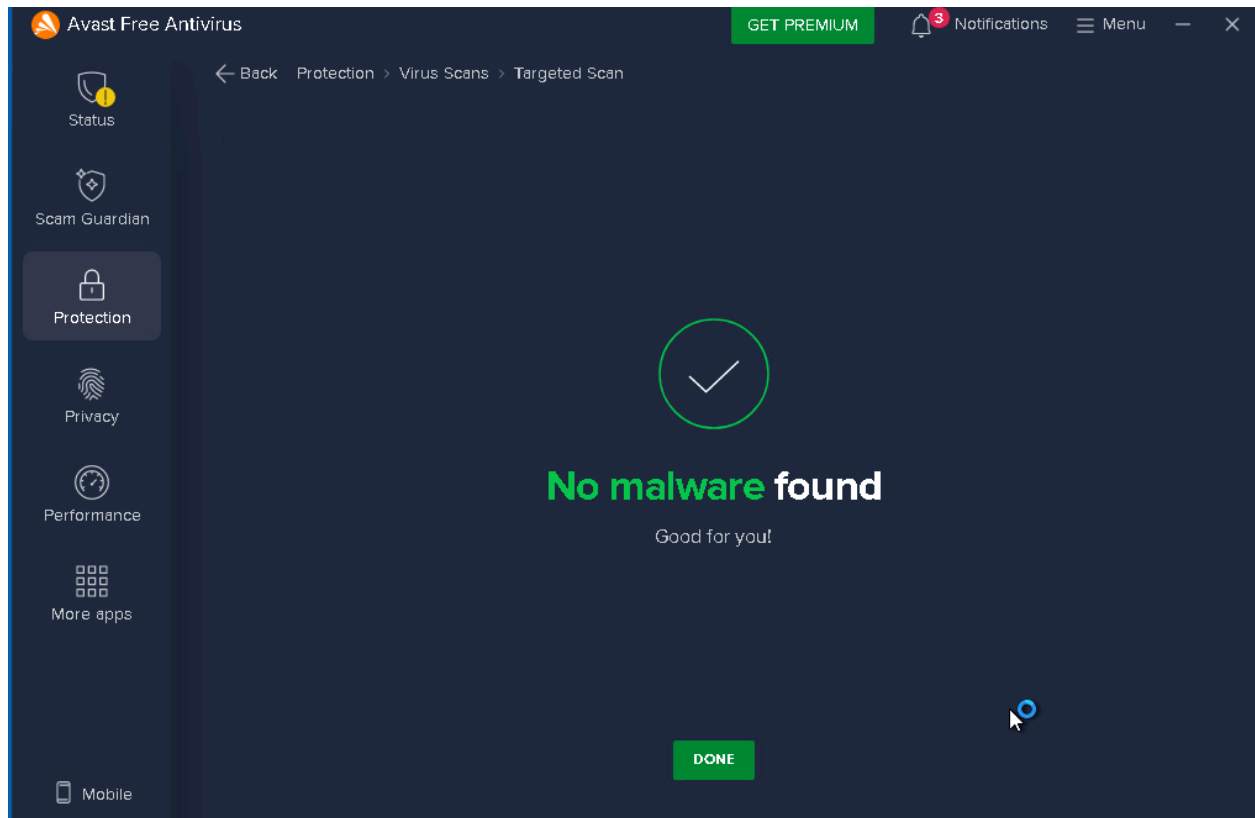
Files: 5

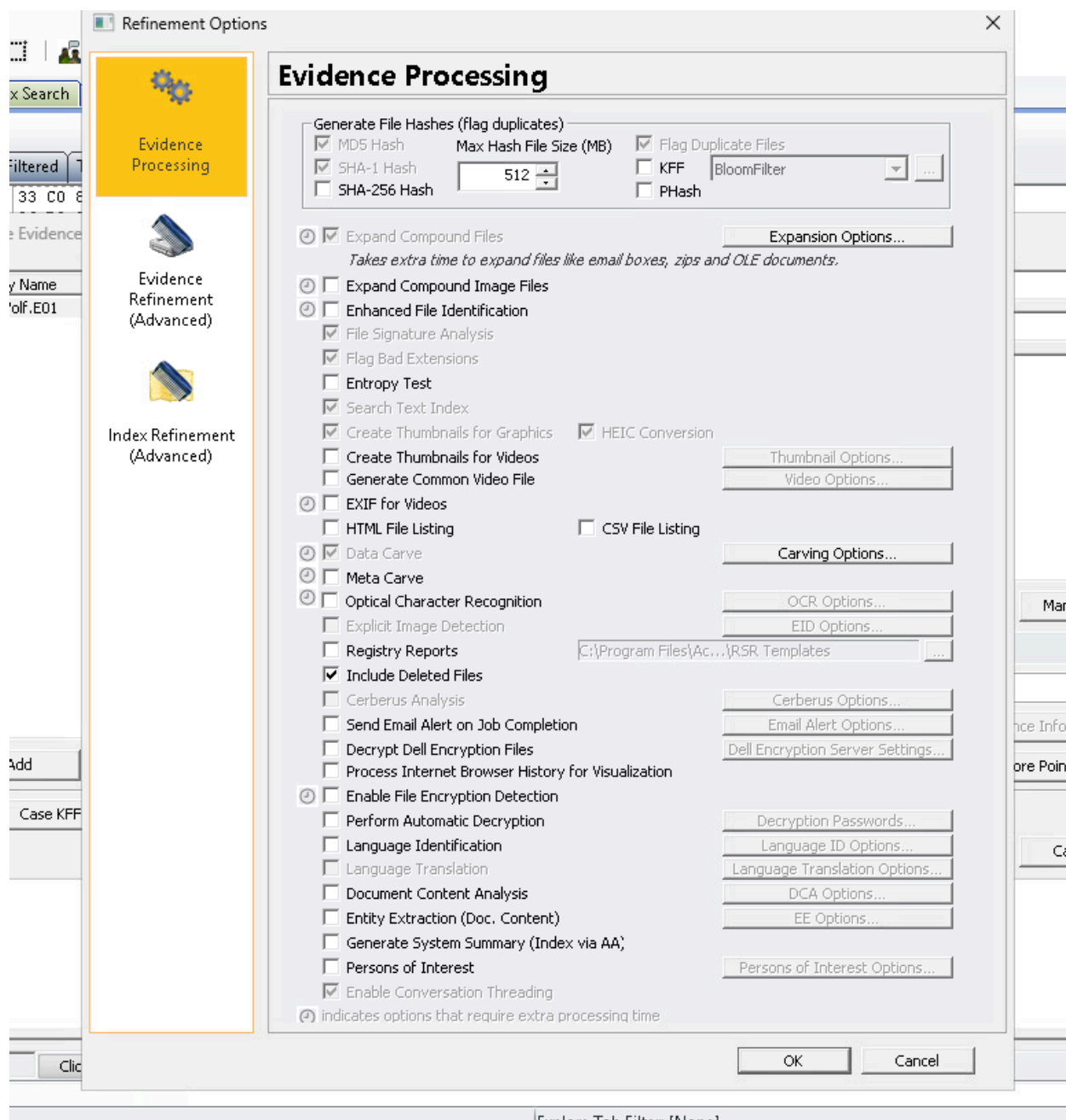
Total: 5

Size: 167889280 B

Checked: 0

Click to update size

Anti-Virus Scan (AVAST)

Evidence processing-

Chain of custody-

CHAIN OF CUSTODY FORM

Bloomsburg Police Department

EVIDENCE CHAIN OF CUSTODY TRACKING FORM

Case Number: 2023-0020 Offense: 001

Submitting Officer: (Name/ID#) Deputy Jones 44279

Victim: Not Applicable

Suspect: Tom W

Date/Time Seized: 04/15/2026 1230 Location of Seizure: Bloomsburg, PA

Description of Evidence		
Item #	Quantity	Description of Item (Model, Serial #, Condition, Marks, Scratches)
001	1	Tom's image stored:d9d54d97378b090b7eac137986a26cdf calculated:d9d54d97378b090b7eac137986a26cdf leslie image stored:8f6c72fa2e31efbe2296332044d86105 Server image stored:520f7fc697b730ac11f43aded0380a7a raid1 3a5e0626f1241d620720225825a5d979 raid 2 a7b327b9a12e7ba9c9087411da72b838 raid 3 df0231540e055e6f606a7ddab7044dc63

Investigate

Item #	Date/Time	Released by (Signature)	Received by (Signature)	Comments/Location
0001	3/1/2026 09:32:04	Dr. Snyder	Jonathan Lunger	Copied image file from L: Drive to local VM C: Drive For analysis